

A social force evacuation model driven by video data

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ABSTRACT

This paper proposes a video data-driven social force model for simulating crowd evacuation. The initialization of pedestrian position, path navigation, and goal selection in the improved social force model was guided by real video data. To initialize pedestrian position and determine path navigation, the distribution of the pedestrians is set according to the real video. We also extracted the trajectories of pedestrian movement from the videos, and these trajectories were stored into a path set to guide the evacuation of pedestrians. Moreover, a fitness function was defined to model the behavior of a pedestrian goal selection. The fitness function could process the evacuation parameters, which were extracted from the video, and consider the degree and distance of exit congestion. Furthermore, we quantified the relationship values among pedestrians, and a new force called “group force” was added to the primary social force model. Pedestrians with close relationship gathered into one group and walked together. To validate the effectiveness of the proposed method, the video data-driven model was applied to simulate campus halls and roads. Simulation results show that the proposed approach is consistent with real-world situations and can assist in analyzing emergency evacuation scenarios.

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1. Introduction

The development of crowd evacuation scenarios has recently become a popular research area [1]. Increasing attention has been paid to determining the safety of people during evacuations. Pedestrian flow in public places, such as supermarkets and subway stations, is mixed, and a small disturbance may cause instability to crowds. Thus, crowd trampling accidents may easily occur. The significance of this issue has been reported [2,3]. A proper evacuation simulation method can predict the danger and save lives during emergencies.

Experiments and simulation modeling are currently the main methods applied to investigate evacuations. Numerous evacuation models, such as social force model (SFM) [4,5], cellular automata models [6,7], fluid dynamic models [8,9], and agent-based models [10], have been established. In most of these models, individuals are treated as autonomous agents, and these individuals can perceive information and decide independently based on their own behavior rules. These models provide a fundamental framework for simulating crowd dynamics. However, finding the correct parameter settings and behavior rules of agents using the traditional models to ensure that the simulation can match the behavior observed in an actual physical system is challenging. Agent behavior rules in agent-based simulations are difficult to derive because the behavior of an agent is affected by his environment and surrounding agents [11].

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Recent studies [32–40] showed that introducing real evacuation data into the traditional model is helpful in analyzing pedestrian movements and making the evacuation more realistic. Several of these models are aimed at learning the rules from video data, and these rules are then used to improve pedestrian movements [12–14]. Consequently, the simulation behavior is apparently similar to the video data. Other data-driven models, such as evolutionary algorithms, are aimed at automatically calibrating model parameters. The potential of data-driven approaches in automatic crowd modeling has been demonstrated. However, macroscopic and microscopic factors should be fully considered in the evacuation process. The macroscopic characteristics and social relationship of pedestrians should be considered. In many practical scenarios, considering only the microscopic features is insufficient to generate the desired emergent crowd dynamics. In the macroscopic aspect, path navigation and goal selection are essential parts of a crowd evacuation simulation model. Moreover, the social relationship of pedestrians affects the evacuation. Pedestrians with close relationship gather into a group and walk together. Thus, we combined the traditional models with the data-driven models in this study to integrate their advantages.

This paper proposes a video data-driven SFM to address the aforementioned issues. The main contributions of this paper are as follows:

- (1) The initialization of the pedestrian position and path navigation was driven by video data. The distribution of pedestrians was set in accordance with a real video. Crowd motion trajectories were also extracted from the video data. An evacuation path set was built after removing repeated paths, and the trajectories were eventually used to guide the crowd evacuation.
- (2) The mechanism of goal selection was driven by video data. A fitness function that considers the degree of congestion and distance of exits was introduced into the original SFM. This fitness function processed parameters, which were extracted from the video; these parameters include the number of pedestrians near the exit and distance to the exits. Then, the fitness of each exit was calculated. Finally, this fitness function modeled the behavior of crowds in goal selection.
- (3) We quantified the relationship values among pedestrians and then added a new force called “group force.” The results after the improvement showed that pedestrians with close relationship gather into one group and walk together.

The rest of this paper is structured as follows. Related works on crowd evacuation models are discussed in Section 2. The data-driven crowd evacuation model is defined based on the improved SFM in Section 3. The top layer models the goal selection and path navigation of the pedestrians, while the bottom layer models the microscopic collision avoidance and relationship among pedestrians. The simulation experiments in school roads and hall settings are presented in Section 4. Then, the simulation results in goal selection, path navigation, and crowd density distribution are analyzed. Finally, the conclusion and future perspectives are presented in Section 5.

2. Related work

The simulation of pedestrian crowds for normal and evacuation situations has been widely examined. Various approaches exist, and each approach focuses on a specific objective. A possible categorization is to distinguish between macroscopic and microscopic models as proposed in Ref. [2]. The dynamic characteristics of crowd flow for macroscopic models are described by average speed, density, and location [15,16]. The detailed behaviors and interactions of the crowd have been disregarded, although the macroscopic model can describe the overall movement trend of the crowd. The mutual influence of individuals is particularly emphasized in microscopic models contrary to macroscopic models. Cellular automata [6,7] are a discrete dynamic crowd movement system that belongs to a microscopic model with grid-based motion decision. The lattice gas model [8,20] is a special case of cellular automata that are based on probability and statistics. The SFM [4,5,21] is also a microscopic model but in continuous time and space. This model introduces a desired force to describe the inner drive of pedestrians to escape, especially under stressful situations.

These traditional models have been improved and designed to make the model realistic. Several early studies have attempted to model the group phenomena [17–19]. Farina et al. [17] added a heading of pedestrians into the dynamic model and introduced a new force to model the behavior of people that walk together as a single group. The model presented by these authors could reproduce the moving behaviors of pedestrians in open spaces and high-density scenarios. Bandini and co-workers [18] presented a research that focused on group and age-driven pedestrian behaviors in an urban crowded scenario. The result showed that dyads walk 30% slower than singles in irregular flows, while elderlies walk 40% slower than adults. Nishinari and co-workers [19] investigated the role of conflicts in pedestrian traffic. A friction parameter μ was introduced into the cellular automaton model for pedestrian dynamics. The simulation results showed that these conflicts are important for an accurate description of the dynamics. The “group force” that we added in our model is determined by the relationship of pedestrians, unlike the abovementioned methods.

The effects of pedestrian vision field [22] and leading mechanism [23–25] have also been considered. Helbing et al. [22] introduced the vision field by defining a term of the socio-psychological force in the original SFM that could reflect the anisotropic character of pedestrian interaction. Several studies have established route sets [26–31] to help in the evacuation. Hoogendoorn et al. [28,30] presented a summary of factors that influence the decision of an individual on the appropriate route selection and suggested numerous schemes by optimizing several established pedestrian-specific utility functions. Ma and Fukuda [29] proposed a route selection model that incorporates hyper path and the generalized extreme values of a network based on link choice models.

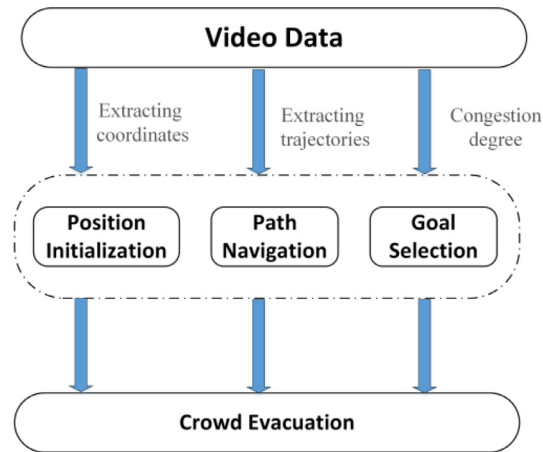


Fig. 1. Data-driven model.

Data-driven crowd modeling has recently gained increasing attention. In Refs. [32–34], the data that were extracted from videos helped in analyzing pedestrian movements. Boltes et al. [32] established two strategies for extracting pedestrian trajectories from stereo recordings in T-junction scenes. One strategy uses a marker for detection, and the other strategy is based on a perspective depth field. Then, these authors obtained and analyzed the density, velocity, and specific flow profiles through the Voronoi method. Georgoudas et al. [33] presented an anticipative system that operates during pedestrian evacuation processes and prevents congestion at the escape points. The system detects and tracks pedestrians, estimates routes, and guides pedestrians toward alternative escape points. Tsiftsis et al. [34] developed an integrated electronic system that allows the dynamic management of congestion and provides a fast evaluation of dynamical circumstances. These authors evaluated the simulation results with real video data and found that the proposed management system can quickly estimate possible routes. Klüpfel et al. [35,36] presented empirical data on an evacuation exercise in football stadiums and primary schools. Three conclusions were drawn: (1) V-like shapes form at the egress from football stadiums; (2) for events with several hundred thousand participants, an evacuation of the complete area is usually inadvisable; and (3) a deviation exists between the egress time predicted by the simulation and the actual time it takes to evacuate the building. Several researchers have attempted to use the detection and tracking method to anticipate and guide pedestrians. In Refs. [37–39], pedestrian trajectories were extracted from videos, and these trajectories were used for path navigation and goal selection. The data-driven crowd modeling frameworks for generating crowd behaviors in this study can match the video data. Was et al. [40] proposed an improved CA model of pedestrian dynamics based on a detailed representation of space combined with a floor field approach. Furthermore, several evacuation tests, video analyses, and analyses of available data have been conducted for various objects, such as universities and stadiums. In Refs. [41,42], optimization algorithms, such as EA and genetic algorithm, were used to calibrate the SFM parameters.

Certain details can still be modified, although many researchers have made improvements. Our model contains improvements in macroscopic and microscopic aspects. In the microscopic aspect, the “group force” was added to the primary SFM to model the group phenomenon. Pedestrians with close relationship could gather and walk as a group. In the macroscopic aspect, we extracted motion trajectories and pedestrian numbers from the video for path guidance and goal selection. The results reflected the effectiveness of our model.

3. Data-driven crowd model

3.1. Model definition

This paper mainly focuses on achieving a realistic crowd evacuation simulation. According to Refs. [28,43], the behaviors of pedestrians mainly depend on the following factors: (1) how pedestrians select their goal; (2) how pedestrians navigate to their goal; (3) how to express the social characteristics of pedestrians, such as gathering and waiting; and (4) how pedestrians avoid collisions with other pedestrians and obstacles. We propose a data-driven crowd evacuation model based on these factors. Fig. 1 illustrates the model structure. In the proposed model, the driven data, such as trajectories and coordinates, were extracted from the video by using Zootracer¹ and an open source video track tool, namely, Tracker.² The coordinates were used to initialize the position of the pedestrians, and movement trajectories were used for path navigation. We defined a fitness function to model the crowd behavior for goal selection. The fitness function received the parameters,

¹ <https://www.microsoft.com/en-us/research/project/zootracer/>.

² <http://physlets.org/tracker/>.

such as path length, movement time, and export congestion degree, which were extracted from the video. Various collision avoidance models, such as RVO and SFM, are available. In 1995, Helbing developed the SFM in which time and space are modeled continuously. The model reproduces well-known self-organizing crowd phenomena, such as “fast is slower” and arch-like blocking in the exit. Thus, we selected the SFM as the bottom component.

3.2. Social force model and its improvement

In the SFM, pedestrians are driven by three forces, namely, desired force $\vec{f}_i^0$, interaction force $\vec{f}_{ij}$ between pedestrians i and j , and interaction force $\vec{f}_{iw}$ between pedestrian i and the walls. Each N pedestrian i of mass m_i aims to move at a certain desired speed v_i^0 in a certain direction $\vec{e}_i^0$ and therefore tends to correspondingly adapt his or her actual velocity v_i^0 with a certain characteristic time τ_i . According to Newton's second law of motion, the corresponding mathematical expression of each pedestrian i is

$$m_i \frac{d\vec{v}_i}{dt} = m_i \frac{v_i^0(t)\vec{e}_i^0 - \vec{v}_i(t)}{\tau_i} + \sum_{j \neq i} \vec{f}_{ij} + \sum_w \vec{f}_{iw}. \quad (1)$$

The second element on the right side of the equation is the interaction forces from other pedestrians (presented in Eq. (2)). $A_i \exp[(r_{ij} - d_{ij})/B_i] \vec{n}_{ij}$ is the repulsive interaction force against each other, and A_i and B_i are constants. d_{ij} denotes the distance between the centers of mass of pedestrians, and $\vec{n}_{ij}$ is the normalized vector that points from pedestrian j to i . The pedestrians touch each other if their distance d_{ij} is shorter than the sum $r_{ij} = (r_i + r_j)$ of their radii, r_i and r_j . In this case, two additional body forces, that is, $k(r_{ij} - d_{ij})\vec{n}_{ij}$ and sliding friction force $\kappa g(r_{ij} - d_{ij})\Delta v_{ji}^t \vec{t}_{ij}$, are added to counteract body compression and impede relative tangential motion.

$$\vec{f}_{ij} = \{A_i \exp[(r_{ij} - d_{ij})/B_i] + kg(r_{ij} - d_{ij})\} \vec{n}_{ij} + \kappa g(r_{ij} - d_{ij}) \Delta v_{ji}^t \vec{t}_{ij} \quad (2)$$

The third element represents the repulsive forces from obstacles (expressed in Eq. (3)). The definitions of parameters are the same as those in Eq. (2). The parameters are $A_i = 2 * 10^3$ N, $B_i = 0.08$ m, $k = 1.2 * 10^5$ kg s⁻², $\kappa = 2.4 * 10^5$ kg m⁻¹ s⁻¹, individual radius: $r = 0.25$ m, individual quality: $m = 80$ kg, and individual expectation velocity: $v_i^0 = 0.8$ m s⁻¹.

$$\vec{f}_{iw} = \{A_i \exp[(r_i - d_{iw})/B_i] + kg(r_i - d_{iw})\} \vec{n}_{iw} - \kappa g(r_i - d_{iw}) (\vec{v}_i \cdot \vec{t}_{iw}) \vec{t}_{iw} \quad (3)$$

In the primary SFM, the pedestrians are initially distributed randomly, which does not correspond to the actual situation of pedestrians' distributions well. Thus, in order to make the simulation of the evacuation results conform to the real video situation from the beginning and match the real video well, we initialized the pedestrians position according to the real video data.

Moreover, all pedestrians are considered equal to one but with different social relationships, such as classmates, friends, and family, in the primary SFM. In the real world, distinct social relationships affect the movement of people. Pedestrians are likely to walk with people who have close or similar relationships with them.

In this study, we performed two improvements. First, we extracted the coordinates of the crowd from a video and used these coordinates to initialize the crowd. Second, we quantified the relationship values among pedestrians, and four relationships were defined in λ :

$$\lambda = \{1, \text{family}; 0.6, \text{friends}; 0.3, \text{colleague}; 0, \text{norelation}\}. \quad (4)$$

A large λ indicates pedestrians with a close relationship. rel_{ij} is a 2D array that records the relationship values of pedestrians i and j .

According to Ref. [24], a leader could gather the pedestrians and improve evacuation efficiency. However, we also considered that several evacuation situations may do not have a real leader. In certain cases, the pedestrians in groups may select to follow the mainstream in the absence of a leader. Therefore, we added a virtual leader to the primary model to display the group phenomenon. Virtual leaders exhibit the largest summation of relationship values among the pedestrians. These virtual leaders guide and gather the crowd and search for the path [44]. Therefore, if a person is the leader of a group, then his desired velocity becomes slower than the normal member. The velocity of the leader is expressed by Eq. (5) as follows:

$$\vec{v}_i^{\text{leader}} = \alpha v_i^0(t) \vec{e}_i^0(t), \quad (5)$$

where $\alpha \in (0 \leq \alpha \leq 1)$, and $\vec{e}_i^0(t)$ is the direction of the destination. If a pedestrian is not the leader of the group, then his desired direction is influenced by the direction of the destination and the position of his leader. This direction is expressed by Eq. (6) as follows:

$$\vec{e}_i^0 = \rho \vec{e}_j^0 + (1 - \rho) \vec{n}_{ij}. \quad (6)$$



(a) Pedestrians with a close relationship were set in the same color and move to the exit.

(b) Arch-like blocking phenomenon near the exit.

Fig. 2. Simulation experiment to validate that the modified SFM can still demonstrate the fundamental crowd characteristics. (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

This paper introduces a new force called “group force” based on the definitions. Pedestrians with close relationship gather together during the evacuation, and this behavior facilitates the determination of pedestrians with the relationship. The mathematical expression of this attractive force is expressed by Eq. (7) as follows:

$$\vec{f}_{ik}^{att} = rel_{ij} C_i \exp[(r_{ij} - d_{ij})/D_i] \vec{n}_{ij}, \quad (7)$$

where $rel_{ij} \in [0,1]$, and rel_{ij} is the weight of the relationship between pedestrians i and j . The weight is significant when a relationship is close. C_i is the strength of the attractive force, D_i is the least safe distance between pedestrians, and $\vec{n}_{ij}$ is the direction of attractive force, $C_i = 2000$ N, and $D_i = 0.5$ m. The new SFM is described as Eq. (8) as follows:

$$m_i \frac{d\vec{v}_i}{dt} = \vec{f}_i^0 + \sum_{i \neq j} \vec{f}_{ij} + \sum_w \vec{f}_{iw} + \sum_{i \neq k} \vec{f}_{ik}^{att}. \quad (8)$$

The improved SFM introduces the group leader and considers the effect of crowd relationship on the evacuation. Thus, the proposed model is more realistic than the traditional model. Furthermore, we performed a simulation experiment to validate that the modified SFM can still extract the fundamental crowd behaviors, such as the arch-like blocking in the exit and the “fast-is-slower” phenomenon, and the results are depicted in Fig. 2.

Fig. 2 presents the 200-pedestrian evacuation results in a single exit room. The width and length of the room are 20 m, and the exit is 1.5 m. Fig. 2(a) demonstrates the pedestrians with a close relationship in the same color and gathers slowly. Fig. 2(b) exhibits the pedestrians who were jammed near the exit and the arch-like blocking at the exit.

Another important feature of the SFM is the “fast-is-slower” phenomenon. In the same evacuation scene, we increased the desired speed of pedestrians from 0.5 m/s to 5 m/s. The simulation results are displayed in Fig. 3. We found that the pedestrian evacuation time decreases first and then increases. The pedestrian evacuation time was at the minimum when the desired velocity increased to 1.5 m/s. However, the collision of pedestrians hindered the evacuation when we continued to increase the desired speed. Thus, the evacuation time began to increase. The above experimental results indicated that the improved SFM can not only simulate the group behaviors of pedestrians but also reproduce the basic behaviors of pedestrians.

3.3. Detection and tracking method

The evacuation scene is presented in Fig. 4(a). The scene has a school hall with a length of 32 m and a width of 21 m, and the room has three exits. The full recording of the pedestrian evacuation of the entire room was difficult by using a single camera. The pedestrian density was large, especially near the exit, and the congestion and overlap were serious. Therefore, to obtain further accurate movement data from pedestrians, we set four cameras in the evacuation scene to record the pedestrian movement trajectories synchronously. Their positions are illustrated in Fig. 4(b). All cameras were mounted at $h = 762$ cm above the floor with the viewing direction that is perpendicular to the floor. The pedestrians with the abovementioned height range can hardly be viewed with occlusion at any time even near the exit. In particular, the cameras, called small ant intelligent camera 2s for recording real-time traffic, have a resolution of 1920×1080 pixels and a frame rate of 16 fps.

The overall goal of the extraction process is the trajectories, which should be as exact as possible from every person. An open source video track analysis tool, namely, “Tracker,” was used in this study. This tool is a free video analysis and

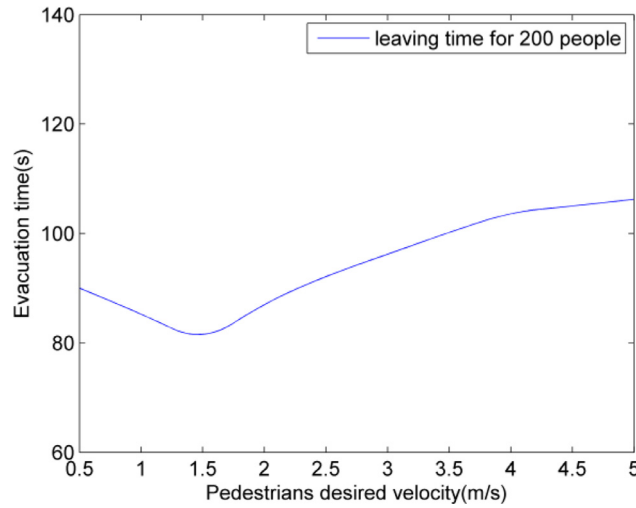
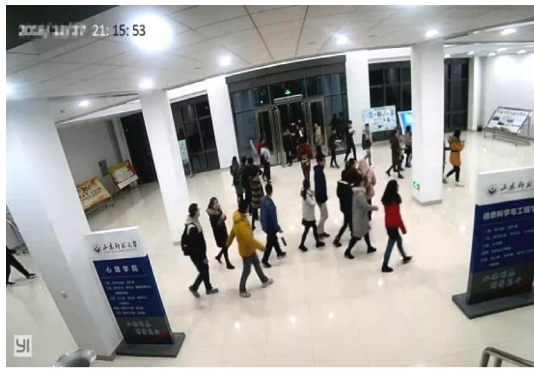
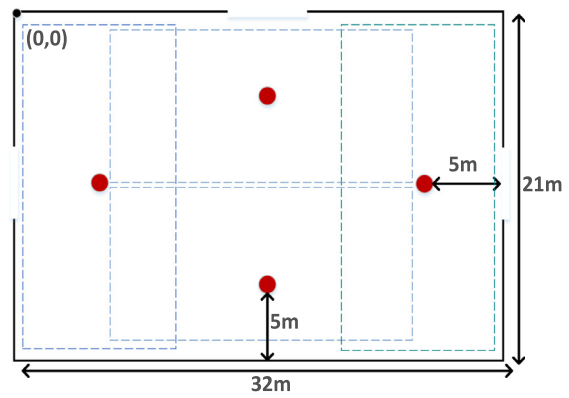


Fig. 3. “Fast-is-slower” phenomenon. The pedestrian evacuation time decreases first and then increases with the increase in the desired velocity of pedestrians.



(a) Evacuation experiment scene from a slanted view.



(b) Positions of the four cameras in the evacuation scene.

Fig. 4. Experiment scene and camera positions.

modeling device, which was built on an open source physics java framework. The Tracker allowed us to extract pedestrian coordinates and trajectories from the video. However, the tool required specifying the starting points of pedestrians manually by clicking on their heads. In particular, the tracker can track multiple targets simultaneously. Several obvious tracking errors can also be corrected manually.

Fig. 5 depicts a snapshot of the tracking result of camera one. In this figure, 14 pedestrians can be viewed, and their movement trajectories are marked with red lines. Specifically, each pedestrian movement coordinate was updated every 0.05 s. These coordinates also allowed us to calculate the real-time distance from the exit to the pedestrians accurately. Other movement characteristics, such as the walk direction and speed, can also be calculated from these coordinates.

A coordinate transformation from the pixel positions to the real positions and an inverse perspective transformation should be performed to calculate the 3D position in the real world. We assumed that the 3D origin coordinate of the real world was set in the top left corner (Fig. 4(b)). According to the camera model,

$$\begin{bmatrix} x_{\text{screen}} \\ y_{\text{screen}} \\ 1 \end{bmatrix} = M_1 M_2 \begin{bmatrix} x_{\text{world}} \\ y_{\text{world}} \\ z_{\text{world}} \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & u_0 & 0 \\ 0 & f_y & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} R & T \\ \vec{0} & 1 \end{bmatrix} \begin{bmatrix} x_{\text{world}} \\ y_{\text{world}} \\ z_{\text{world}} \\ 1 \end{bmatrix}. \quad (9)$$

Each coordinate in the 3D real world corresponds to a point in the image. Their conversion formula is expressed in Eq. (9). M_1 is the 3×4 intrinsic matrix that was determined by using the internal parameters of the camera, (f_x, f_y) is the focal point, and (u_0, v_0) is the center of projection. M_2 consists of the rotation matrix R and translation matrix T . In this



Fig. 5. Snapshot of the tracking result of camera one. (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

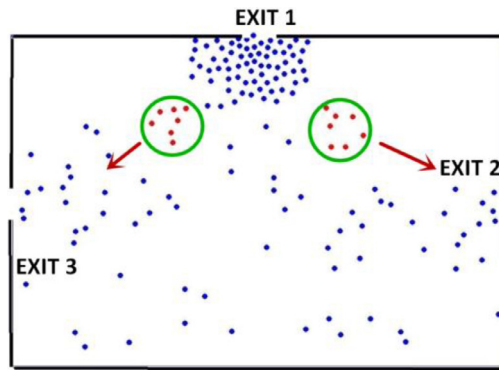


Fig. 6. Goal selection of pedestrians considering the congestion. (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

study, the four cameras we used had the same focal length and resolution. These cameras were all mounted at $h = 762$ cm, with the viewing direction perpendicular to the floor, thereby indicating that these cameras had the same rotation matrix. The only difference was their translation matrix T ; the matrix can be determined by the positions of the cameras. The constant parameters of the cameras are as follows: $f_x = 3765.83$, $f_y = 3969.79$, $u_0 = 947.65$, and $v_0 = 495.48$. Finally, the 3D real-world coordinates could be calculated from the pixel coordinates with the OpenCV computer vision library.

3.4. Goal selection

In the primary SFM, the pedestrians select the nearest exit to evacuate. However, an actual video was analyzed, and the results showed that people considered the degree of congestion and their position when selecting a goal [45,46].

Fig. 6 demonstrates the distribution of a population during a real evacuation. We extracted the coordinates from the video and then used MATLAB to describe the video. This scene contains three exits. The two groups of red particles selected exits 2 and 3 during the evacuation, although these groups were close to exit 1. Their behaviors were due to the extended waiting time if they would leave from exit 1 than evacuate from the other exits. Thus, we introduced a fitness function to the primary SFM to model the behaviors of pedestrians in selecting their goal. The fitness function parameters, such as length of the path and individual numbers, can be obtained from the videos, and the mathematical expression of this function is expressed by Eq. (10) as follows:

$$F(O_i) = c_1 \sqrt{\sum_{i+1 \leq j} \sum_{j=1}^d (O_{ij} - O_{i+1,j})^2} + c_2 \frac{\text{count}[O_i]}{J_{O_i}}, \quad (10)$$

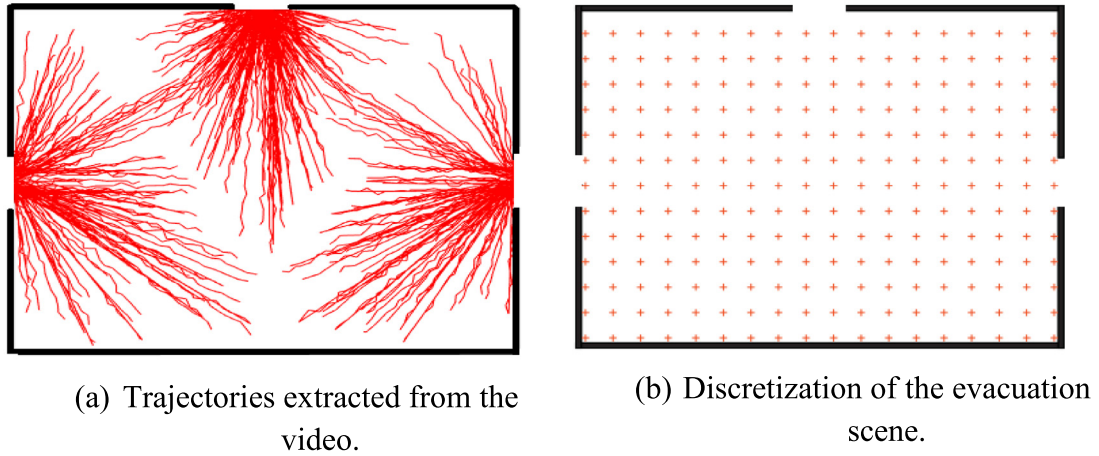


Fig. 7. Evacuation trajectories and scene.

where d is the dimension of solution space, $\sqrt{\sum_{i+1 \leq j}^d (O_{ij} - O_{i+1,j})^2}$ is the length of individual movement from O_i to O_j , and $\text{count}[O_i]$ is the number of individuals who selected O_i as their exit. J_{O_i} is the evacuation speed of O_i , and c_1 and c_2 are the weight parameters ($c_1 = 0.5, c_2 = 0.5$). The algorithm is described as follows:

The fitness function is used to model the selection of the target in a real scenario by calculating the distance and degree of congestion. If numerous individuals select exit O_i as their goal and the evacuation speed is slow, then the fitness values are small. The individual will select another target to avoid the blockage of a large crowd. We model the psychological state of the escape of an individual based on this kind of target selection mechanism.

3.5. Path navigation

Path navigation plays an important role in crowd evacuation. In this study, we assumed that the group leaders can navigate the path, and other pedestrians follow the group leader. The navigation path we used was extracted from the real videos of pedestrian evacuation. The repeated path and transforming trajectories from “pixel coordinate system” to “world coordinate system” are stored in the path set after removing these trajectories.

The data that we extracted from the videos are trajectories, and each trajectory is represented by a series of vectors: $T = \langle e_1, e_2, \dots, e_n \rangle$, where $e_i = (x_i, y_i)$ represents the position of the i point of the trajectory, and n represents the total number of trajectory points. The extracted trajectories are short and repetitive.

We recorded a video of pedestrians that walk in the school hall and extracted the trajectories from the video. Fig. 7(a) plots the trajectories used in the simulation studies. Numerous repetitive trajectories exist, and these trajectories are difficult to use. According to Ref. [31], an effective means of addressing this problem is to discretize the evacuation scene and route. The evacuation scene is discretized into a series of cells similar to cellular automation. These cells are represented by a center point, or a key point, as displayed in Fig. 7(b). If the entire region is divided into $H \times W$ grids, then the region can be represented by matrix g , where $g_{i,j}$ is the $\text{grid}(i, j)$ center coordinates. Each trajectory that is extracted from the video will pass through the adjacent grids. The discretization process of an evacuation route is the process of finding the nearest key point from matrix g to replace the point on the route. If e_i is a point in the original route, then replaced key point k_i is calculated by Eq. (11) as follows:

$$k_i = \min \|e_i - k_i\|, k_i \in g. \quad (11)$$

where $\|e_i - k_i\|$ is the Euclidean distance between e_i and k_i . The proposed method demonstrates flexibility in generating complex moving patterns of pedestrians and can reduce the duplication of trajectories. Fig. 8 exhibits the trajectories in which duplicated paths have been removed. The available evacuation routes cover the entire scenario space. The group leader at every position in the scenario can select a route as their navigation path, and the group members can follow their leader.

4. Simulation

To verify the effectiveness of the proposed data-driven SFM, the framework was applied to simulate the crowd behaviors in different scenes. We designed several contrast experiments in the school roads and hall. In experiment one, we especially selected the scene in a campus road and used an unmanned aerial vehicle to record videos of pedestrian evacuation for approximately 20 min. Then, the distribution of crowd walking paths was used as an evaluation criterion to analyze the

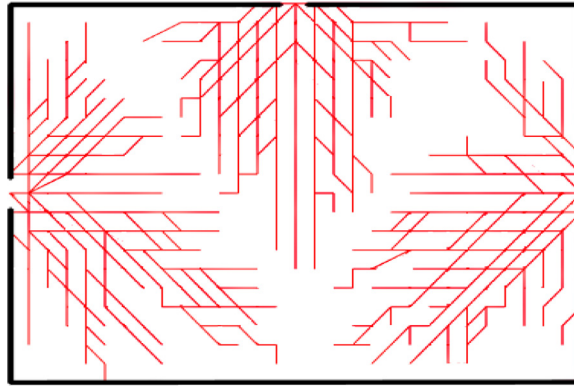


Fig. 8. Simplified crowd motion trajectories by discretizing the evacuation scene.

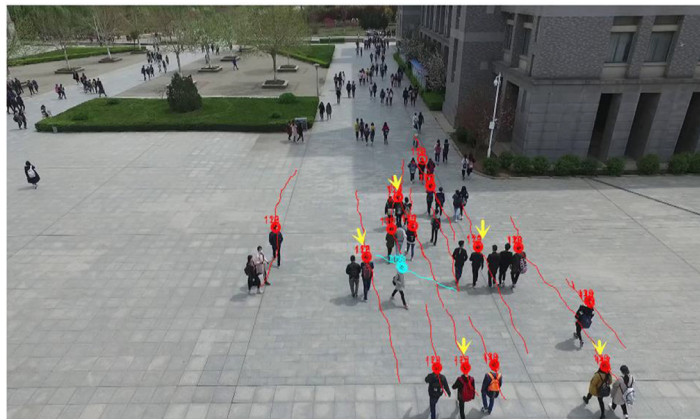


Fig. 9. Extracted trajectories from the evacuation scenarios on a school road. (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

effectiveness and validity of our models. In experiment two, we selected the school hall as our simulation scene. The results show that our model is practical in the aspect of export choice. Particularly, the evacuation time of our model in a multiple-exit room was consistent with the real evacuation time, and the original SFM consumed more time than the real time.

4.1. Simulation on a school road

The scene in experiment one is illustrated in Fig. 9. We deleted the irrelevant scene during modeling and only kept the area of pedestrian concentration. A $100\text{ m} \times 30\text{ m}$ 3D scene model was built. Then, we extracted the trajectory from the video. The virtual leaders were marked with yellow arrows. The red paths in Fig. 9 are the typical paths that we extracted. These paths included the characteristics of pedestrian movement, such as the direction of movement, trajectory, and movement speed. However, these trajectories were in the pixel coordinate system. Thus, these trajectories were transformed into world coordinates. These paths were used after the transition to guide the pedestrian evacuation in our SFM.

We used the original SFM and our approach to plan the crowd evacuation routes. The evacuation effects are depicted in Fig. 10. Specifically, Fig. 10(a) demonstrates the original SFM simulation result. The pedestrians in the evacuation scenarios were discretely distributed because the effect of the pedestrian relationship was disregarded. Fig. 10(b) illustrates the results of our proposed simulation method. The virtual leaders were marked with blue arrows. Pedestrians with a close relationship walked together and kept a distance from strange particles.

Fig. 11 exhibits the density distributions of crowds in the SFM and the proposed method. Our simulation results were more realistic than the original SFM.

A significant feature can be observed in Fig. 11(a), that is, the pedestrian distribution is discrete, and the pedestrians are distant from each other. This phenomenon is mainly due to the repulsive forces from other pedestrians that cause the particles to avoid each other. Moreover, the relative position among the particles will remain unchanged when the resultant force among the pedestrians reaches the equilibrium state. Such scenario is infeasible in the real world. In Fig. 11(b), the “group force” was added to the SFM. Pedestrians with a close relationship gathered into one group and moved together. Their distribution was similar to the video data.

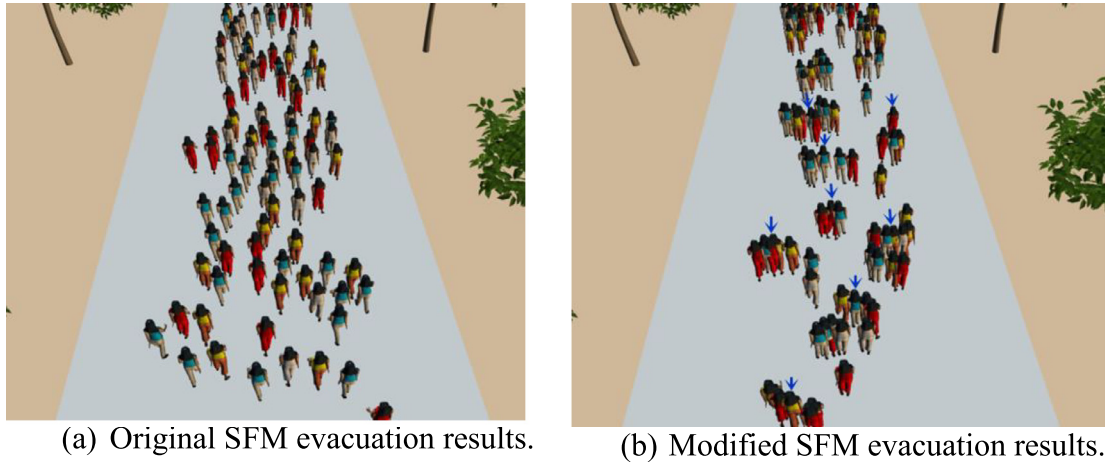


Fig. 10. Comparison of crowd evacuation simulations on a school road. (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

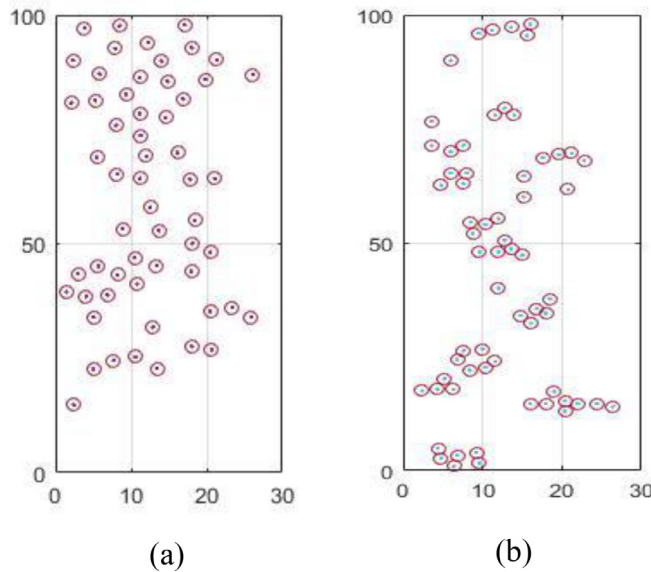


Fig. 11. Comparison of the density distributions of the crowds.

Fig. 12 displays the trajectories of the crowd in the video and those generated by the SFM and the proposed method. These trajectories were generated by the pedestrians in four groups. The crowd evacuated from the same starting point, but the movement trajectories significantly differed. The red trajectories were extracted from the video data, and the blue trajectories showed our data-driven model simulation results. The green trajectories illustrated the SFM simulation result. Our simulation results were nearly consistent with the real trajectories and even contained an “S” shape compared with the SFM.

This characteristic is primarily because the trajectories extracted from the video in our data-driven model were used to guide the pedestrian evacuation. The group leader walked along the trajectories, and the group members followed the leader. Thus, the movement trajectories were similar. However, the trajectory of the original SFM was remotely distinct from the real trajectory due to insufficient guidance. Moreover, the SFM evacuation trajectory was nearly a straight line, and the details of the movement, such as direction changing, were ignored. Such exclusion is due to the force in the SFM can easily reach a stable state with sparse population distribution. The relative position of crowd movement is unchanged when the state stabilizes.

The abovementioned comparison results that we presented are mainly qualitative. Therefore, a quantitative analysis was necessary to validate the authenticity of the improved SFM simulation results.

We calculated the cumulative error distance between the two models. We calculated the two Euclidean distances l_i^1 and l_i^2 in every second. l_i^1 is the distance between the real and the modified SFM motion coordinates in the i th second. Similarly,

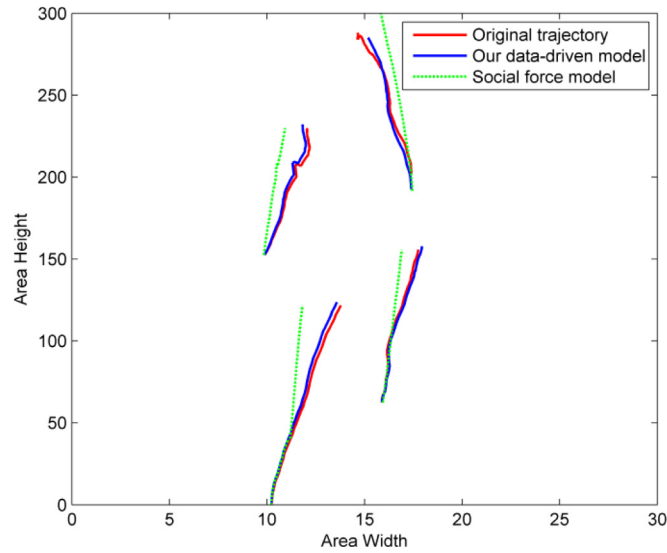
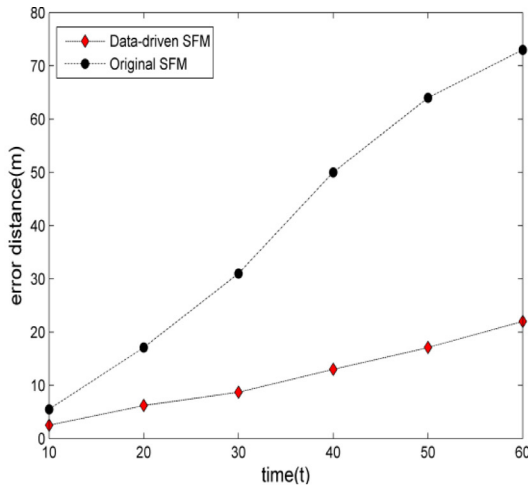
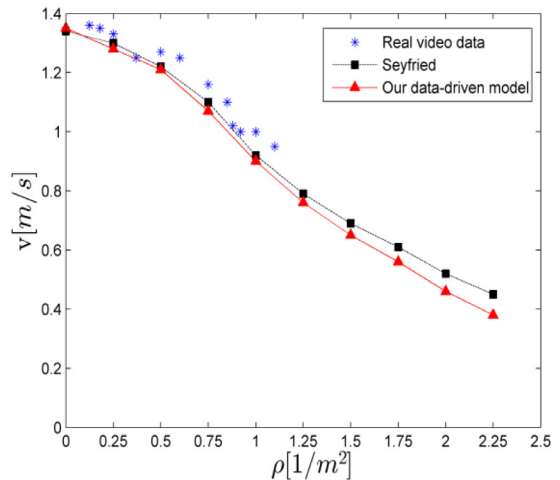


Fig. 12. Comparison of the motion trajectories. (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)



(a) Comparison of the cumulative error distance.



(b) Comparison of density–velocity.

Fig. 13. Quantitative comparison.

l_i^2 is the distance between the real and the original SFM motion coordinates in the i th second. Therefore, $L_n^1 = \sum_{i=1}^n l_i^1$ and $L_n^2 = \sum_{i=1}^n l_i^2$ can be considered the cumulative error distance of the two models within n seconds. The model is accurate when the cumulative error distance is small. We counted the average error distance of 220 people over 60 s. The contrast results are displayed in Fig. 13(a). The modified SFM cumulative error distance is smaller than the original SFM, thereby denoting that the modified SFM matches the video data better than the original SFM.

Another quantitative analysis of velocity–density is also presented in this paper. First, we calculated the average speed and density of pedestrians in relation to the real video to verify the effectiveness of the extraction method. In Fig. 13(b), our statistical results basically match the results of Ref. [47], which also verifies the correctness of the extraction method, when the pedestrian density is within the range of 0–1.1. Furthermore, the data-driven SFM can produce diagrams that are similar to the fundamental diagrams reported in the literature [47]. However, the crowd mean velocity of our model descended rapidly for the density of pedestrians (up to 1.25 p/m²) because the difficulty of people that gather in groups increased, and the leader waited longer when the pedestrian density increased. Finally, the velocity of our model dropped quickly. This result is also consistent with actual situations.

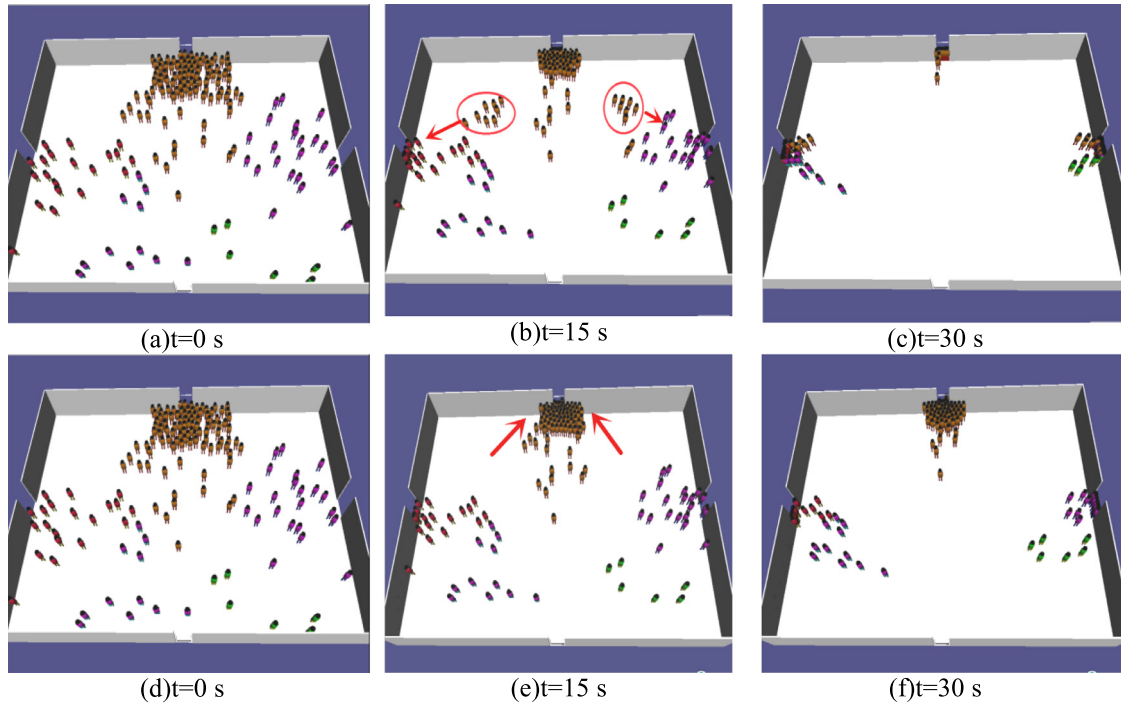


Fig. 14. Snapshots of the crowd evacuation simulation in the school hall. In (a) and (d), the crowd is initialized in the same position. In (b) and (e), the pedestrians in red circle exhibit a different movement direction. In (c) and (f), the evacuation time of the two models is different. The pedestrians in our model nearly left the room. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Comparison of evacuation times for the three models.

Pedestrian number	Real evacuation time (s)	Data-driven SFM time (s)	Original SFM time (s)
100	20.68	21.26	22.58
200	33.6	35.58	41.04
300	46	48.02	57.26

4.2. Simulation in school hall

Fig. 14 illustrates the simulation scene, which includes three exits and one entrance, and each exit has the same width of 1.5 m. The proposed data-driven model and original SFM were used to evacuate 200 pedestrians in the same initialization condition. We analyzed the simulation results from the target selection and the time consumption. The results show that the proposed model simulation results were realistic.

We analyzed the goal selection of the two models. Figs. 14(a)–(c) depicts the snapshots of the modified SFM of pedestrian evacuation. Fig. 14(a) illustrates that numerous pedestrians gather near the exits. Fig. 14(b) displays that the two groups of pedestrians encountered serious congestion near the exit, and thus the pedestrians in red circles selected the left and right exits rather than the exits in front of these pedestrians. This phenomenon is consistent with the real situation. This behavior is a normal characteristic of pedestrians because such movement saves time. Figs. 14(d)–(f) demonstrates the simulation results of the original SFM. Fig. 14(d) exhibits that the pedestrians are initialized at the same location. Specifically, Fig. 14(e) presents that pedestrians select the nearest exits even if the exit is seriously congested. This assumption is inconsistent with reality.

Another important difference between the two models is that our model consumes minimal evacuation time. Specifically, Fig. 14(c) illustrates that the pedestrians are nearly evacuated from the room at evacuation time $t = 30$ s. In Fig. 14(f), numerous pedestrians in the original SFM are still waiting to be evacuated. A detailed comparison of the evacuation time for the two models is summarized in Table 1. Our data-driven SFM evacuation time is less than that of the original model and is more consistent with the actual situation.

The comparison of the original SFM and the proposed model indicates two differences. First, the target selection of pedestrians in our model is realistic. The congestion degree and distance of the exits are considered. Second, the evacuation time is shorter in our model than in the original model and closer to the real evacuation data.

Algorithm 1 Crowd goal selection algorithm.**Input:**The individual number near O_i : $count[O_i]$;Exit evacuation efficiency: J_{O_i} ;The distance from O_i to O_j : $\sqrt{\sum_{i+1 \leq j}^d (O_{ij} - O_{i+1,j})^2}$.**Output:**

Each exit fitness values.

1. For each exit, calculate the fitness values:

$$F(O_i) = c_1 \sqrt{\sum_{i+1 \leq j}^d (O_{ij} - O_{i+1,j})^2} + c_2 \frac{count[O_i]}{J_{O_i}} \quad (c_1 = 0.5, c_2 = 0.5)$$

2. Compare the fitness values and return a minimum values exit.

5. Conclusion and future work

A data-driven SFM was proposed to simulate realistic crowd behaviors. In the macroscopic aspect, path navigation and goal selection were driven by video data. For path navigation, we extracted the trajectories of crowd movement from the videos, and these trajectories were stored in a path set to guide the crowd evacuation. For goal selection, we introduced a fitness function to determine the congestion degree and pedestrian goal selection. The fitness function parameters, such as the pedestrian number and movement speed, were extracted from the video data. The pedestrian goal selection from this fitness function considers the congestion degree and distance of the exits. In the microscopic aspect, we improved the primary SFM, that is, a new “group force” that considers individual relationships, was added to the SFM. The improved SFM can be implemented to prevent pedestrian collision and study the gathering behaviors. The proposed framework was finally used to simulate crowd behaviors in school scenarios. The comparison of the simulation results and real evacuation videos indicate that the proposed framework is effective in generating the desired crowd dynamics and provides a promising prediction performance.

Two important conclusions can be drawn, and these conclusions are consistent with real-world situations. First, the nearest exits are frequently not the fastest exits. In an emergency, an individual should fully use all exits to improve the evacuation efficiency. Second, pedestrians move in groups during evacuation given the social relationship among pedestrians.

The trajectory extraction algorithm we used is suitable for minimal pedestrian overlaps and requires marking the starting points of pedestrians manually. For real-time applications, the detection and tracking method should be improved. In our future work, we will modify the tracking algorithm and build a real-time crowd evacuation simulation system. We will predict the flow of people and avoid congestion by collecting real-time camera video data.

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